Zero-Shot vs Fine-Tuned: A Comprehensive Evaluation Across Encoder, Decoder, and Transformer Architectures for Single-Class Text Classification

Authors: Helen Lu, Monica Martin, Kirthi Shanbhag

GitHub Repo Link

Abstract:

This project explores the trade-offs between zero-shot LLM and fine-tuned transformer models for single class text classification on real-world, imbalanced, domain-specific datasets. While zero-shot models like Mistral and GPT-4 enable rapid adaptation without labeled data, fine-tuned models such as BERT often achieve higher accuracy and efficiency when sufficient training data exists. We systematically evaluate BERT, Mistral, and T5 model architectures to provide practical guidance on model performance, scalability, and deployment in data-limited and resource-sensitive settings.

I. Introduction & Background:

Recent NLP advances highlight a central trade-off between zero-shot LLMs and fine-tuned transformer encoders for text classification (Zhang et. al., 2025). Zero-shot models like GPT-4 generalize well and adapt quickly without labeled data, making them attractive in low-resource settings. Yet growing evidence shows that fine-tuned encoder models often achieve higher accuracy, greater stability, and better efficiency when even modest task-specific data is available (Bucher et. al., 2024). Large-scale comparisons across many datasets and languages further reveal that zero-shot LLMs perform solidly on sentiment tasks but are frequently outperformed on non-sentiment, multi-class, or multilingual classification (Vajjala et. al., 2025). Synthetic LLM-generated training data can help, but typically does not exceed the performance of a well-tuned encoder model. Overall, task-aware model selection remains essential.

This project builds on these insights by comparing supervised fine-tuning and zero-shot approaches across BERT, T5, and GPT-4 for single-class text classification. Given the dominance and utility of highly accurate single-class text classification models, our goal is to provide practical guidance on how to achieve performant models with few resources.

II. Methods:

2.1 Data

For this project we utilized three publicly available text datasets within the healthcare domain: HuggingFace Medical Abstracts, Kaggle Mental Health, and HuggingFace Drug Review. For each dataset we removed rows with missing data and conducted exploratory data analysis for class imbalance. All datasets exhibit varying degrees of class imbalance. To address this, we applied different balancing strategies:

1. For the Medical Abstracts dataset, presence of multiclass classification led us to utilize a random tie-break methodology to choose a label. This was to preserve statistical fidelity. However, the

provided, pre-split test data is incongruent with the train dataset in terms of nuance and complexity which was reflected in lower accuracy scores across both baseline and finetuned models. We used stratified sampling to preserve similarities in train, validation, and test data populations.

2. For the Drug Review dataset, the original review ratings were collapsed into a 1–5 rating bucket, resulting in a noticeably imbalanced distribution across classes. Before modeling, the dataset was cleaned by removing extra spaces, special characters, HTML tags, quotation marks, and unnecessary whitespace. To provide richer context to the model, three important fields—*drug review*, *drug name*, and *condition*—were concatenated into a single input column separated by the '|' symbol.
3. For the Mental Health dataset we collapsed the diagnostic labels to four clinical categories based on DSM-5-TR groupings: Depressive Spectrum, Anxiety, Stress, Bipolar, Personality, and Normal. We utilized hybrid random resampling post-consolidation to handle class imbalance.Mainly:
    1. Target Size: The count of the "Normal" class was set as the target n_samples.
    2. Random Undersampling (Majority Classes): Samples were removed at random without replacement until the class size matched 16,040.
    3. Random Oversampling (Minority Classes): Samples were randomly duplicated with replacement until the class size reached 16,040.

Each technique was selected based on empirical performance for each dataset. This design enabled a consistent, cross-dataset and cross-model evaluation framework for assessing zero-shot and fine-tuned model performances.

2.2 Baseline Models

We selected Logistic Regression as our baseline because it provides a simple, efficient, and reasonable starting point for comparison, and it consistently delivered stable accuracy across all three datasets. This allowed us to meaningfully contrast its performance with our finetuned and LLM models. The baselines for each dataset were: 0.54 F1 for Medical Abstracts, 0.55 F1 for Drug Review, and 0.93 F1 for Mental Health.

2.3 Modeling:

For each of our architecture types, we altered hyperparameters in order to optimize performance. This was for us to compare the capabilities of each architecture. For our evaluation metrics we used the classic metrics including Precision, Recall, and F1-score, in order to compare model performance across architectures.

2.3.1 T5:

We chose the T5 base model because its text-to-text architecture allows it to handle diverse classification tasks with strong generalization, making it well-suited for our three datasets. To ensure consistency and comparability across experiments, we trained all three datasets using the same set of hyperparameters and a slightly higher number of epochs and table show the highest accuracy achieved. We also experimented with several imbalance-handling strategies, including up sampling, down sampling, and weighted sampling; however, these techniques did not yield noticeable improvements, and the overall results remained the same. Based on these observations, we ultimately selected a simpler configuration without any sampling strategies, as it provided stable performance while reducing unnecessary complexity.

Dataset	Epochs	Train Loss	Val Loss	Accuracy	Precision	Recall	F1 Macro	F1 Weighted
Medical Abstract	4	0.32140	0.41102	0.69760	0.68660	0.71142	0.69637	0.69637
Drug Review	3	0.42640	0.45470	0.64357	0.49882	0.49067	0.49004	0.49004
Mental Health	5	0.03080	0.04776	0.97111	0.97124	0.97112	0.97104	0.97104
Max Token Length = 128, Learning Rate = 5e-5								

2.3.2 BERT:

For our BERT-like model we selected PubMedBERT-base (Mezzeti 2023), built on the standard BERT-base architecture and fine-tuned using the PubMed baseline, a biomedical literature dataset. Optimizing for performance, we chose varying hyperparameters for each of our three datasets:

Dataset	Epochs	Training Loss	Validation Loss	Accuracy	F1 Macro	F1 Weighted
Medical Abstract	3	0.44190	0.79601	0.73253	0.73157	0.72666
Drug Review	5	0.44680	1.152537	0.66342	0.54291	0.66468
Mental Health	2	0.07440	0.15118	0.97153	0.97149	0.97149
Max Token Length = 256 / 512, Learning Rate = 2e-5						

2.3.3 Mistral:

We selected the **Mistral 7B** model for its contextual reasoning and adaptability through LoRA fine-tuning, though it remained computationally intensive, requiring approximately **2–5 hours** per two-epoch training cycle on an A100 GPU. Mistral achieved stable convergence and competitive performance across datasets. On the *Drug Review* dataset, the results indicate balanced generalization despite class imbalance. Performance was moderate on *Medical Abstracts* and highest on *Mental Health*.

Dataset	Epochs	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1 Macro	F1 Weighted
Medical Abstract	2	1.0737	1.334088	0.566	0.6310	0.5657	0.55	0.4849
Drug Review	2	1.3536	1.560493	0.7165	0.6957	0.7165	0.5737	0.7034
Mental Health	2	0.9534	1.1719	0.9186	0.9238	0.9186	0.9184	0.9184
Max Token Length = 50, Learning Rate = 2e-4								

III. Results and Discussion

Our final test weighted F1 scores are as follows:

Dataset	Model		
	PubMedBERT	T5	Mistral
	F1 Score	F1 Score	F1 Score
Medical Abstracts	0.63	0.64	0.60
Drug Reviews	0.66	0.40	0.72
Mental Health	0.97	0.97	0.92

PubMedBERT: This model tended to overfit, particularly on the Medical Abstracts dataset given the differences in complexity and nuance between the training and test sets. After trying a few approaches such as a custom weighted loss function, it became clear that a simpler, lighter model outperforms one that is highly curated and trained on the training and validation sets. After experimenting with different hyperparameters and loss functions with little to no improvement, we concluded that the large gap between validation and test metrics was due to the variability, complexity, and nuance differences in the data.

The PubMedBERT model saw the greatest F1 scores across two of the three datasets, and only slightly lower on the remaining dataset. This suggests that BERT handles concise, domain-specific, and well-labeled text particularly well, capturing fine-grained semantic distinctions. Training and validation losses are generally aligned, indicating that the model did not severely overfit and maintained consistent behavior across the three datasets.

The Drug Review dataset incongruencies between the training and test populations continued to cause the model to underperform. This behavior is consistent across the three models we tried on this dataset.

T5 base: We noticed limited improvement after fine-tuning. Because our data was noisy and highly subjective, the model has limited signal to learn from, which reduces the performance gains achievable through fine-tuning. Even after experimenting with various imbalance-handling strategies such as up sampling, down sampling, and weighted sampling, the improvements remained marginal. This suggests that the inherent variability and ambiguity in user reviews constrain the extent to which fine-tuning can boost accuracy.

T5 performed strongly on two of the three datasets, excelling in tasks with clear lexical patterns like Medical Abstracts and Mental Health. It produced stable results with a single hyperparameter setup, generalizing well without tuning. Compared to PubMedBERT, it achieved higher F1 on Medical Abstracts and matched performance on Mental Health.

Like the PubMedBERT model, T5 struggled on the highly imbalanced, subjective Drug Review dataset, frequently misclassifying minority classes. Even with upsampling, downsampling, and class weighting, gains were minimal, suggesting that noise, sentiment variability, and inconsistent cues in user-generated reviews limit the model's ability to learn clear class boundaries.

Mistral :The Mistral 7B model demonstrated robust contextual reasoning and strong generalization across all three datasets, with particularly high performance on the *Mental Health* corpus (F1 ≈ 0.92). Despite its computational intensity, requiring 2–5 hours per two-epoch fine-tuning cycle on an A100 GPU, its parameter-efficient LoRA configuration enabled efficient adaptation without severe overfitting. On the *Drug Review* dataset, Mistral achieved balanced performance (**accuracy = 0.7165**, **precision = 0.6957**, **recall = 0.7165**, **weighted F1 = 0.7034**), indicating stable learning despite class imbalance and subjective variability. While its performance on *Medical Abstracts* was moderate (**F1 Weighted = 0.4849**), Mistral excelled in tasks requiring nuanced interpretation of emotional or domain-specific language. These results suggest that Mistral's transformer architecture effectively captures semantic depth and contextual dependencies, though at a significantly higher computational cost compared to lighter models such as T5 or PubMedBERT.

Model Comparison

Across all three datasets, the models exhibited clear strengths tied to their underlying architectures and pretraining domains. **PubMedBERT** achieved the most consistent performance, leading on Medical Abstracts and Mental Health tasks due to its biomedical specialization and strong representation of concise, domain-specific text. While generally well-aligned between training and validation performance, it showed mild overfitting on *Medical Abstracts*, likely due to data complexity and limited variability.

T5 demonstrated strong generalization across structured text, performing particularly well on Medical Abstracts and Mental Health (F1 ≈ 0.97). However, it struggled with the highly subjective and imbalanced *Drug Review* dataset, where multiple imbalance-handling strategies yielded minimal improvement. This suggests that linguistic ambiguity and sentiment variability in user-generated reviews limited T5's ability to form consistent class boundaries.

Mistral 7B achieved competitive results across all datasets, showing especially high contextual accuracy on Mental Health (F1 ≈ 0.92) and balanced performance on Drug Review (accuracy = 0.7165, weighted F1 = 0.7034). Its LoRA fine-tuning enabled efficient adaptation without overfitting, though performance on Medical Abstracts was moderate (F1 Weighted = 0.4849). While Mistral's transformer architecture captured deeper semantic dependencies than T5 or PubMedBERT, this came at a significantly higher computational cost, highlighting a trade-off between contextual depth and efficiency.

Conclusion

Overall, the results highlight a trade-off between efficiency and contextual understanding across models. **PubMedBERT** and **T5** offered strong, reliable performance on structured and domain-specific text while remaining computationally efficient, making them suitable for most classification tasks. **Mistral 7B**, although more resource-intensive, delivered superior comprehension of subtle variations in tone and sentiment, particularly in the Mental Health dataset. These findings suggest that model selection should align with the complexity and interpretive demands of the target domain, lighter models for efficiency and generalization, and larger transformer architectures like Mistral when deeper semantic understanding is required.

Future Work

Future work should explore more advanced imbalance-handling techniques such as focal loss, label smoothing, or constrained decoding to stabilize predictions for low-frequency classes and reduce bias toward dominant categories. Incorporating data augmentation or synthetic text generation using large language models (LLMs) could also expand underrepresented classes and improve generalization. Building on current results, testing larger or instruction-tuned models such as Flan-T5-large or Mistral-Instruct may enhance the model's ability to capture richer semantic and sentiment patterns.

Beyond architectural improvements, future efforts could reframe the Drug Review classification task as an aspect-based or multi-label problem, allowing the model to recognize multiple sentiment dimensions (e.g., efficacy, side effects, satisfaction) within a single review. Further evaluation could include attention visualization to better interpret model focus in sensitive domains like mental health. Finally, introducing domain adaptation or cross-dataset transfer learning could strengthen robustness and improve generalizability across applications.

Appendix I: Sources

1. Hang Zhao, et. al. “Advancing Single and Multi-task Text Classification through Large Language Model Fine-tuning.” In: *arXiv:2412.08587* (2024).

2. Hauzenberger, L. (2024, January 15). “*Multilabel Classification using Mistral-7B on a single GPU with quantization and LoRA*.” Medium. https://medium.com/@lukas.hauzenberger/multilabel-classification-using-mistral-7b-on-a-single-gpu-with-quantization-and-lora-8f848b5237f3

3. Junyan Zhang et. al. “Do BERT-Like Bidirectional Models Still Perform Better on Text Classification in the Era of LLMs?” In: *arXiv:2505.18215* (2025).

4. Martin Juan José Bucher et. al. “Fine-Tuned 'Small' LLMs (Still) Significantly Outperform Zero-Shot Generative AI Models in Text Classification.” In: *arXiv:2406.08660* (2024).

5. Sowmya Vajjala, et. al. “Text Classification in the LLM Era - Where do we stand?” In: *arXiv:2502.11830* (2025).

Appendix II: Author Contributions

Helen Lu : Led the exploratory data analysis for the **Medical Abstract** dataset, including cleaning and preparing the data for downstream experiments. Implemented baseline models using logistic regression and fine-tuned an encoder-based model (BERT). Contributed major sections of the written report, including methodology and results for the Medical Abstract experiments. Additionally, updated and refined the presentation slides for the Medical Abstract and BERT model sections.

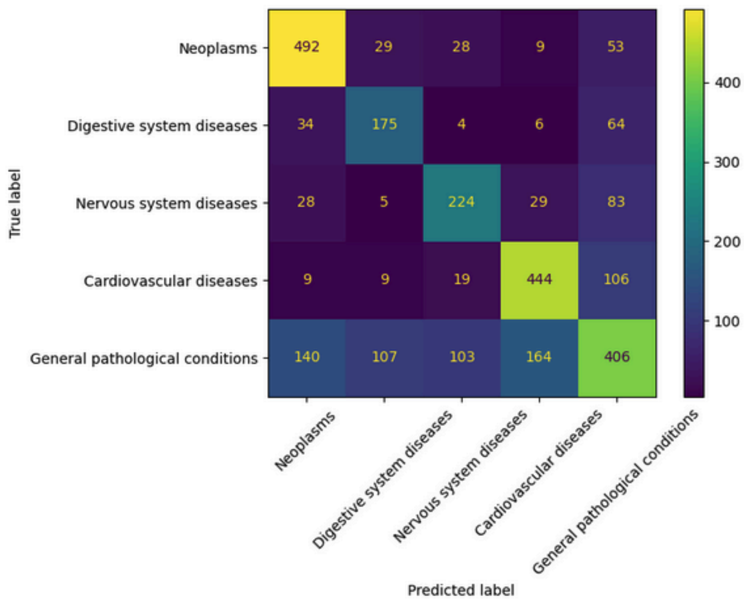
Kirthi Shanbhag : Identified and evaluated multiple candidate datasets for the project, performing initial feasibility studies. Conducted EDA and data cleaning for the **Drug Review** dataset. Developed baseline models, performed zero-shot analysis, and fine-tuned multiple LLMs. Carried out inference and validation using decoder-based models such as **T5** and encoder-based models such as **ClinicalBERT**, ultimately selecting **T5** as the primary model for downstream tasks. Authored the T5 methodology, results, and abstract sections of the report. Person 2 also created and updated the slide deck for the T5 model, baseline comparisons, and dataset overview.

Monica Martin : Led the work on the **Mental Health EDS** dataset. Exploratory data analysis (EDA) was first conducted to examine class distribution, text length, and linguistic variability. Using guidance from the **DSM-5** diagnostic framework, overlapping or semantically related categories were consolidated into four primary diagnostic groups, *Normal*, *Anxiety/Stress*, *Bipolar/Personality*, and *Depressive Spectrum,* to ensure clinically meaningful and non-redundant classification. To mitigate imbalance, a **hybrid up- and down-sampling** approach was applied, resulting in approximately 16,000 samples per class. Focused on developing, fine-tuning, and evaluating the **Mistral 7B** model. The **Mistral 7B** transformer was selected for its ability to model long and context-rich text, and was fine-tuned using **LoRA parameter-efficient adaptation** in a causal language modeling, instruction/response format.

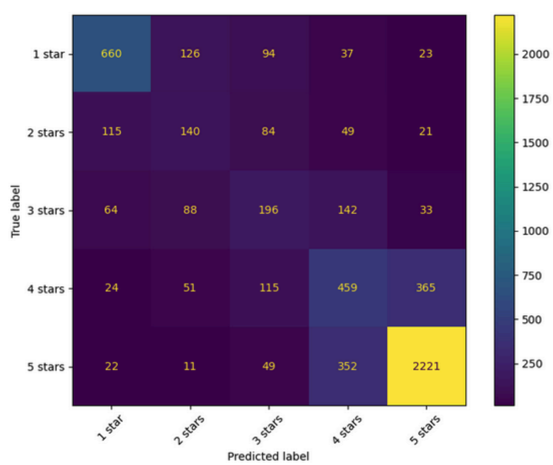
Appendix III: Model Details

PubMedBERT Confusion Matrices

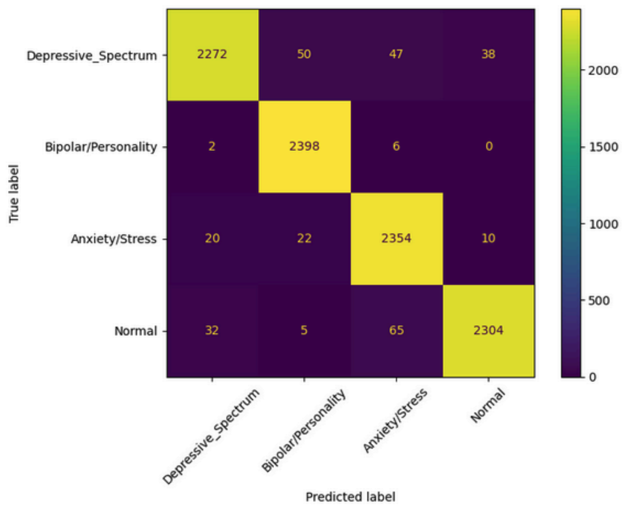
Drug Review:



Medical Abstracts:

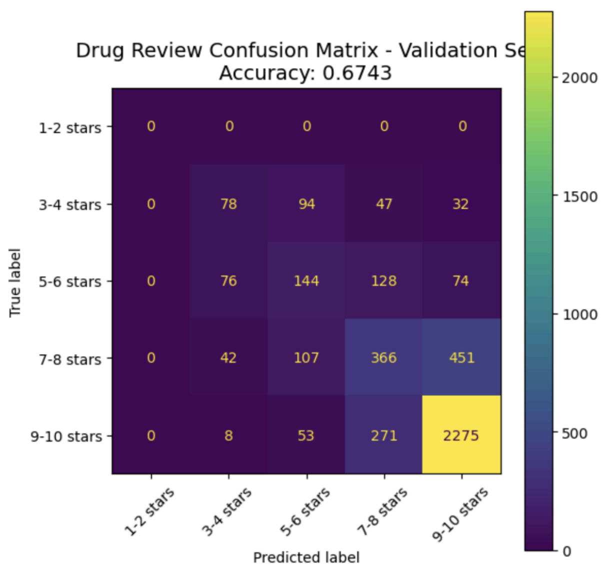


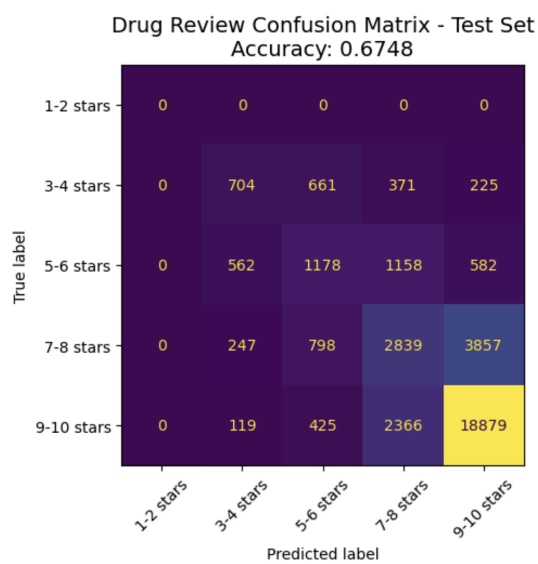
Mental Health:



T5-base Confusion Matrices:

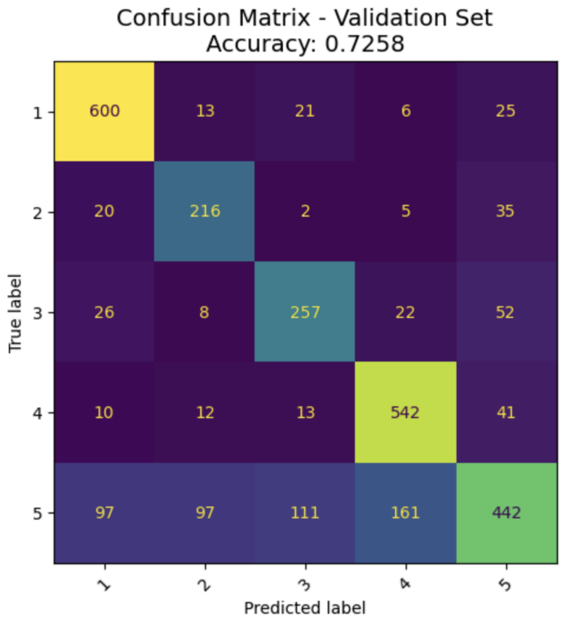
Drug Review:

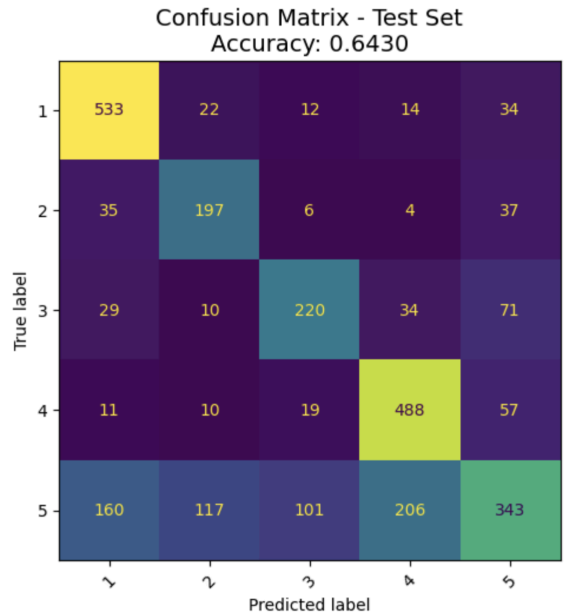




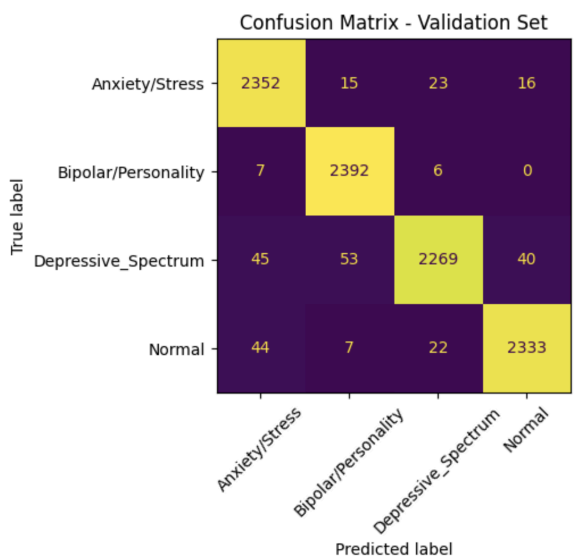
* Please refer to the legend from this diagram for all the others.

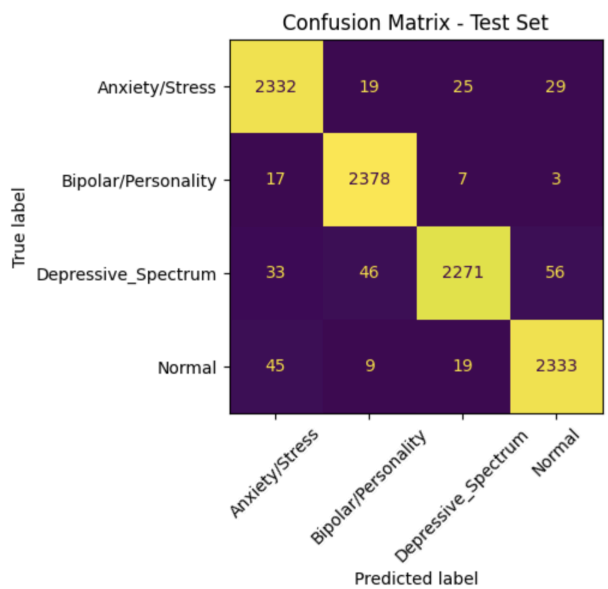
Medical Abstract:





Mental Health:



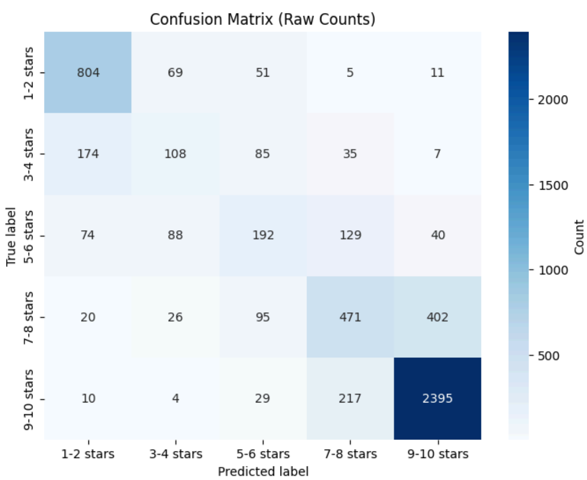


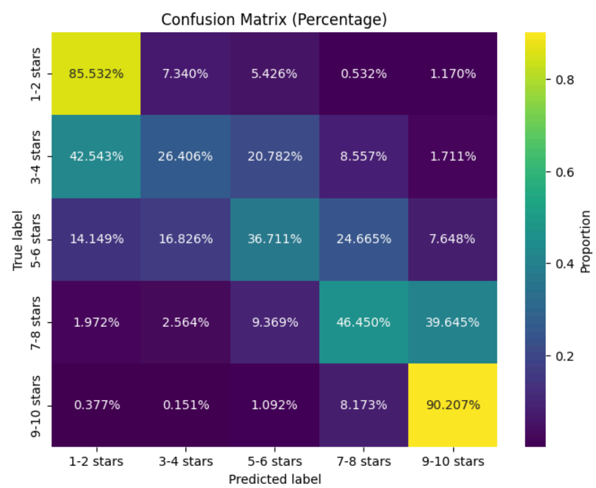
Model 3: Mistral 7B

Note: Mistral 7B's large-scale architecture made each training session computationally expensive, taking several hours per run even with LoRA optimization

Drug Review:

The Mistral 7B model performed strongly on the Drug Review dataset, accurately identifying nuanced clinical and emotional language in user feedback. It achieved balanced precision and recall across categories but required significant computational resources and long runtimes, making it powerful yet costly compared to lighter models like logistic regression. We saw Weaker results for mid-range reviews (3–8 stars), where language tends to be more neutral or ambiguous.





Medical Abstract:

The high recall for *Neoplasms* (0.9574) coupled with low precision (0.5056) suggests that the model is biased toward over-identifying this class, likely due to underlying dataset imbalance or strong class-specific lexical patterns. In contrast, *General Pathological Conditions* exhibit extremely low recall (0.0663), indicating underrepresentation or ambiguity in textual cues. The confusion matrix reveals frequent misclassification of other disease categories as *Neoplasms*. Overall, the Mistral 7B model achieved moderate performance (macro F1 = 0.55), but class imbalance remains a limiting factor for recall and precision parity across medical categories.

Mental Health:
